EFS (ELASTIC FILE SYSTEM)

Amazon Elastic File System is a scalable network file storage service used with Linux EC2 instances. It allows multiple EC2 instances to access the same file system at the same time.

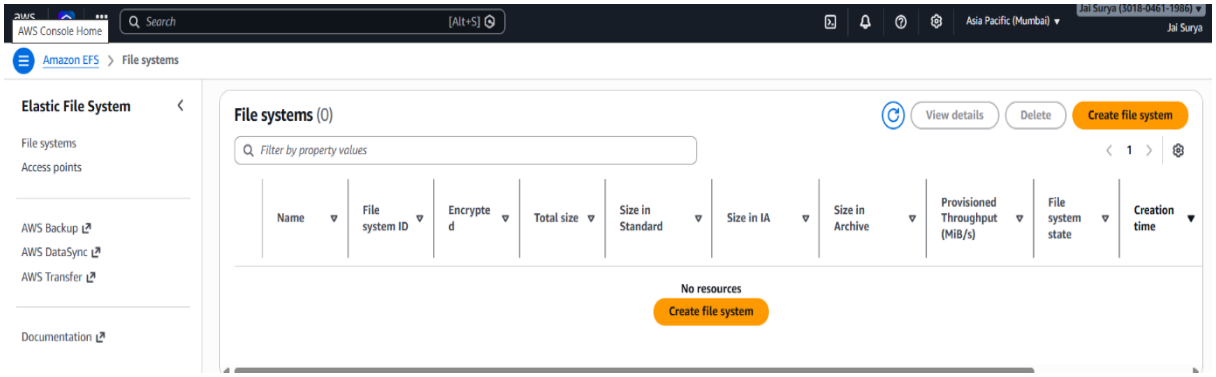
Key Features

- Fully managed by AWS
- Automatically scales storage
- Shared file system
- Accessible from multiple EC2 instances
- Uses NFS protocol

HANDS-ON STEPS (SCREENSHOTS)

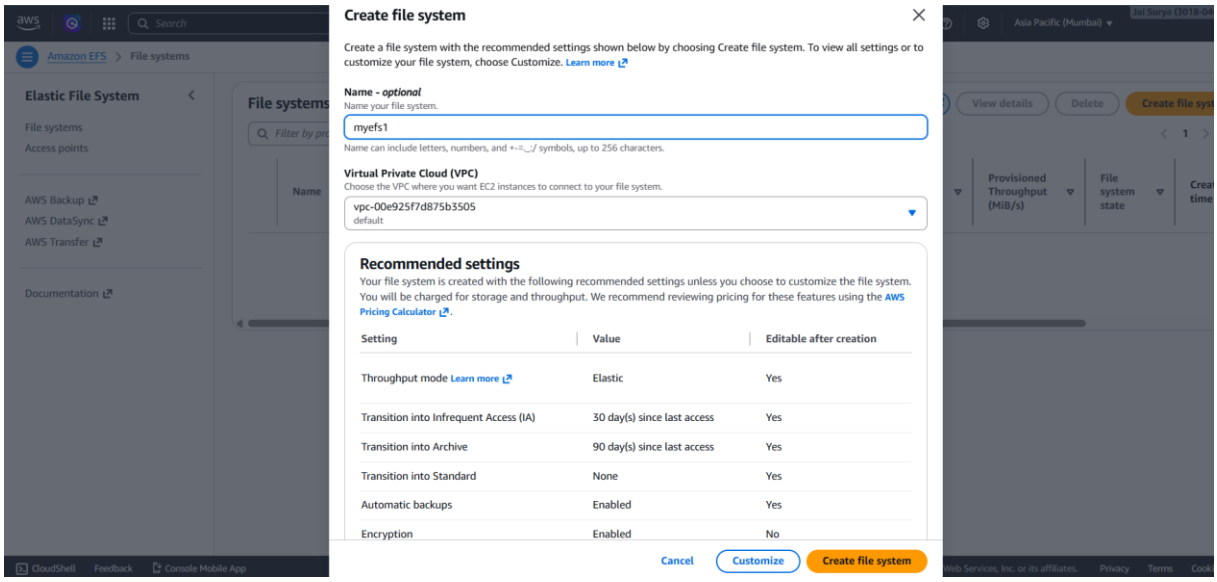
Screenshot: Click Create File System

1. Login to the AWS Console.
2. In the search bar, type EFS.
3. Open Elastic File System.
4. Click Create file system.



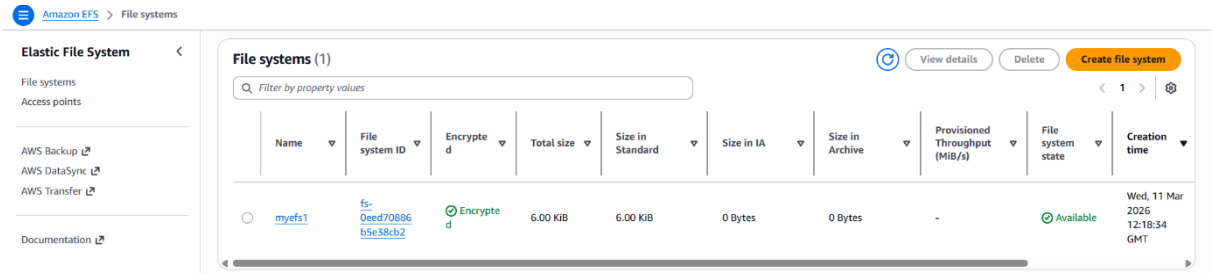
Screenshot: Create File System

1. Go to Amazon Elastic File System in the AWS Console.
2. Enter the File System Name.
3. Review the settings (VPC and default configuration).
4. Click Create to create the file system.



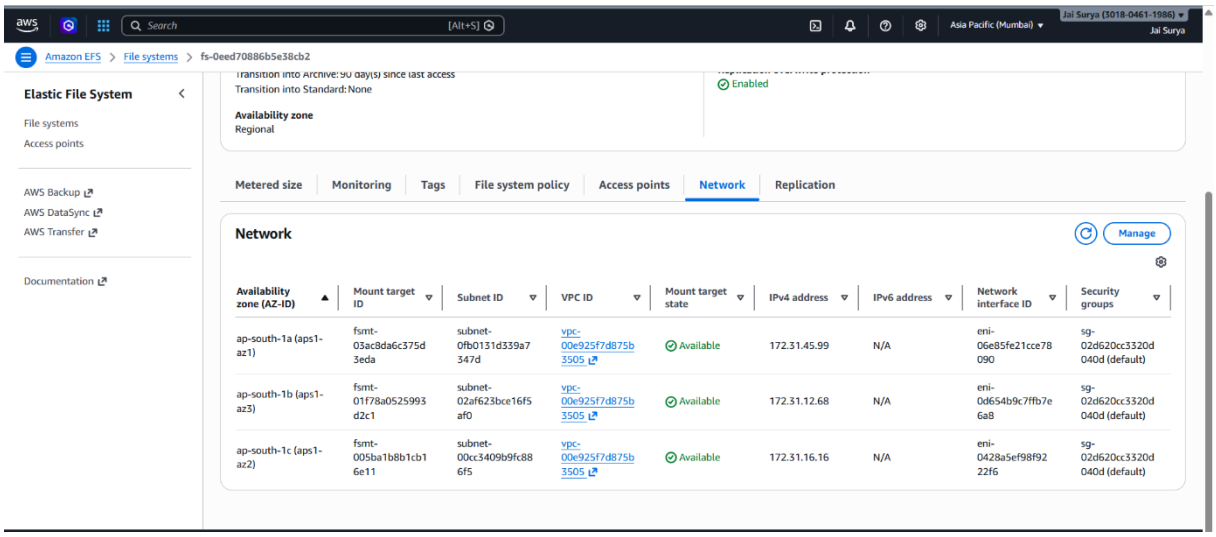
Screenshot: File System Created Successfully

1. After clicking Create, the Amazon Elastic File System will be created.
2. The File System ID and Name will be displayed in the list.
3. The Status will show Available.



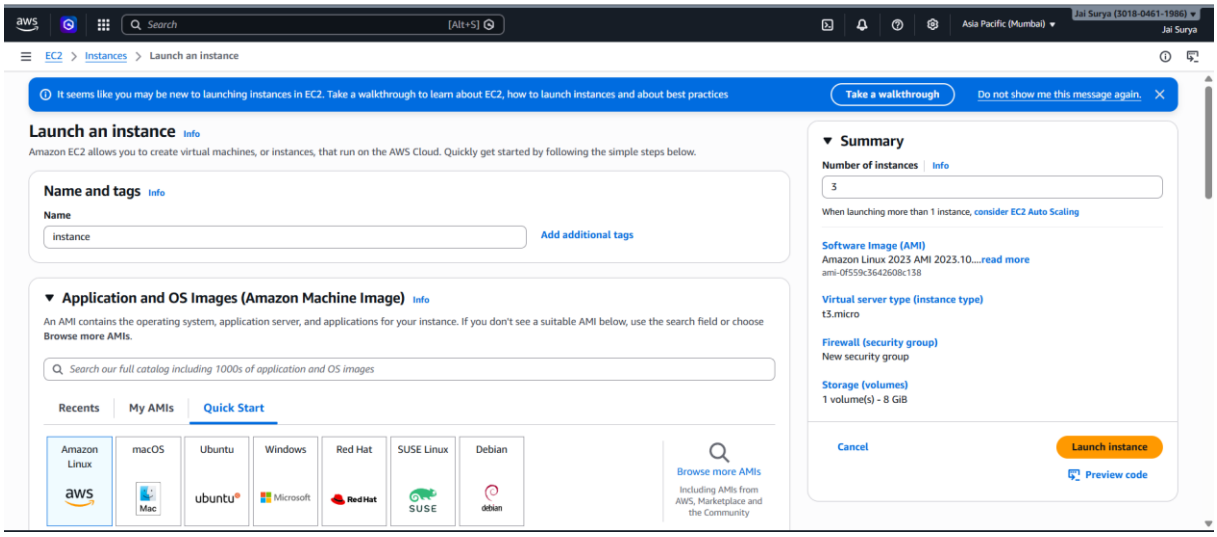
Screenshot: Check Network Settings in EFS

1. Open the created Amazon Elastic File System.
2. Click the Network tab.
3. Verify the Mount Target Status is Available.
4. Check the attached Security Group.



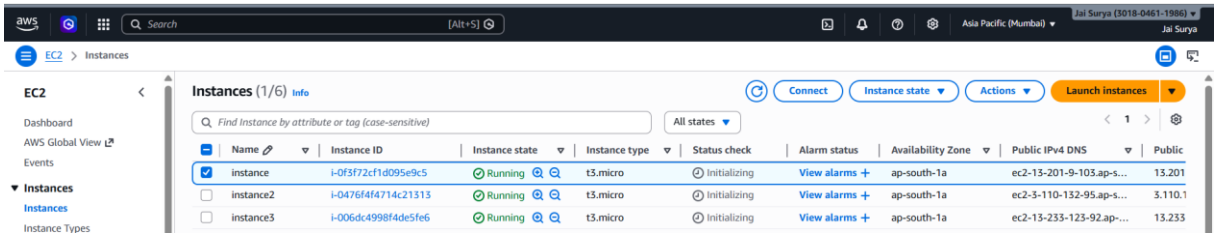
Screenshot: Create EC2 Instances

1. Click Launch Instance.
2. Enter the Instance Name.
3. Select Linux (Amazon Linux) as the operating system.
4. Set the Number of Instances = 3.
5. Select the Key Pair for secure login.
6. Review the settings and click Launch Instance.



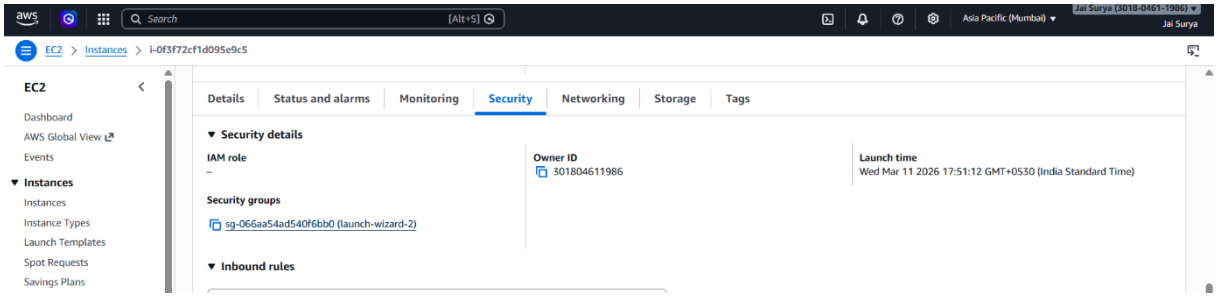
Screenshot: EC2 Instances Created Successfully

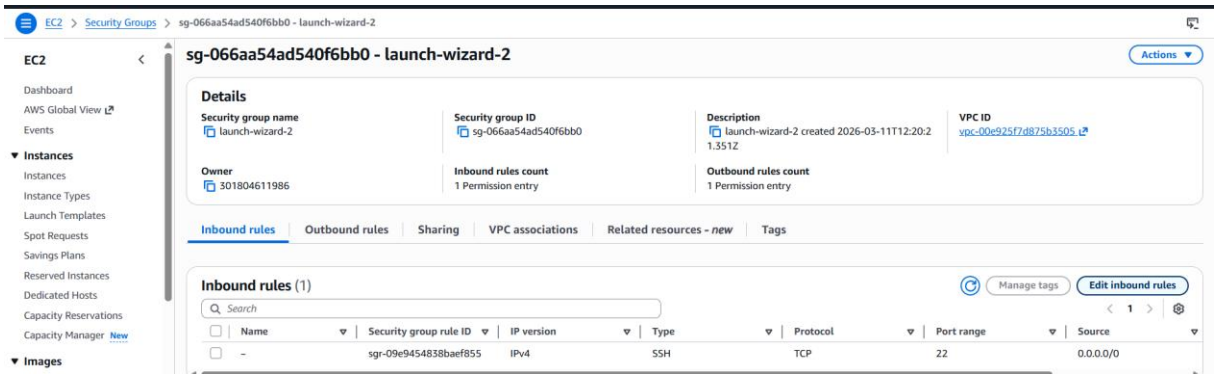
1. After clicking Launch Instance, the Amazon EC2 instances will be created.
2. You can see the three Linux instances in the Instances list.
3. Check that the Instance State shows Running.



Screenshot: Edit Security Group Inbound Rules

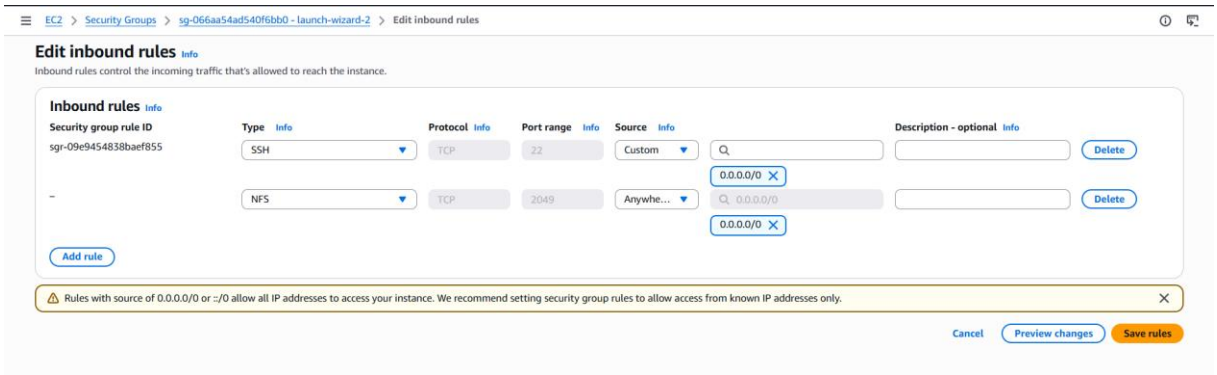
1. Select the EC2 instance in Amazon EC2.
2. Go to the Security tab.
3. Click the Security Group ID.
4. Select Edit Inbound Rules.





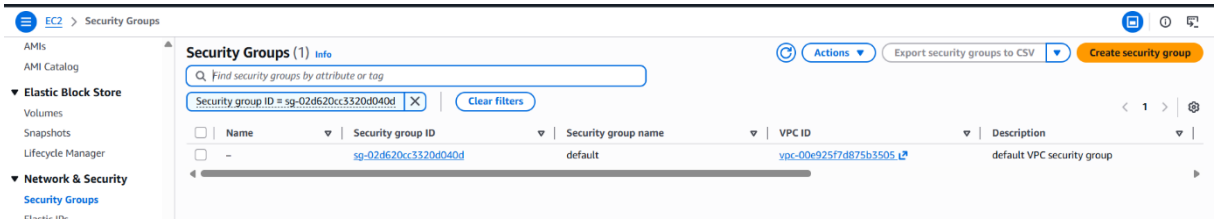
Screenshot: Add NFS Rule

1. In the Security Group Inbound Rules, click Add Rule.
2. Select Type: NFS.
3. Set the Source as the required security group or Anywhere (for testing).
4. Click Save Rules.



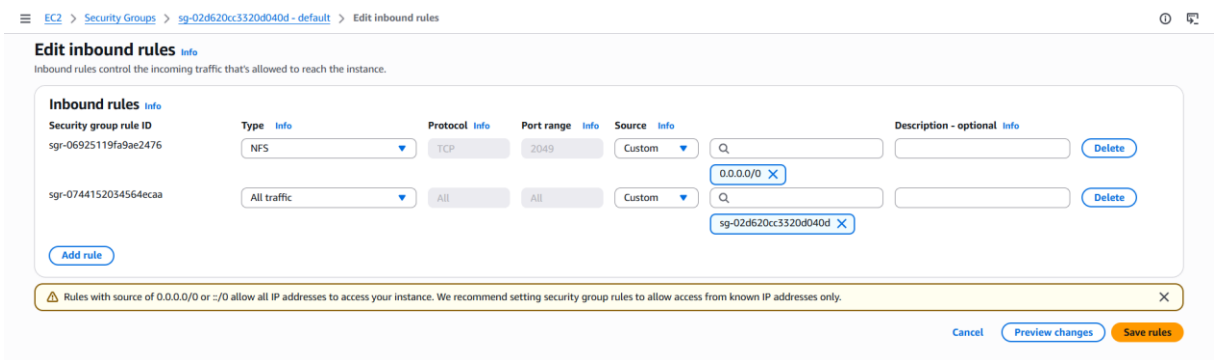
Screenshot: Add EC2 Security Group in EFS Network

1. Go to Amazon Elastic File System.
2. Open the created EFS File System.
3. Go to the Network tab.
4. In Security Groups, click Manage or Edit.
5. Paste the Security Group ID copied



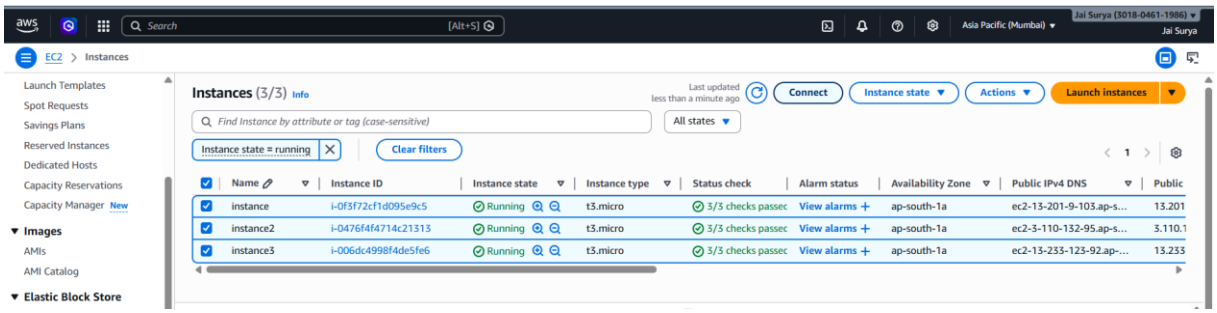
Screenshot: Check NFS Rule in Security Group

1. Go to Amazon EC2 in the AWS Console.
2. Click Security Groups in the left panel.
3. Paste the Security Group ID copied earlier in the search bar.
4. Open the Security Group.
5. Go to Inbound Rules.
6. Verify that the NFS (Port 2049) rule is added and saved.



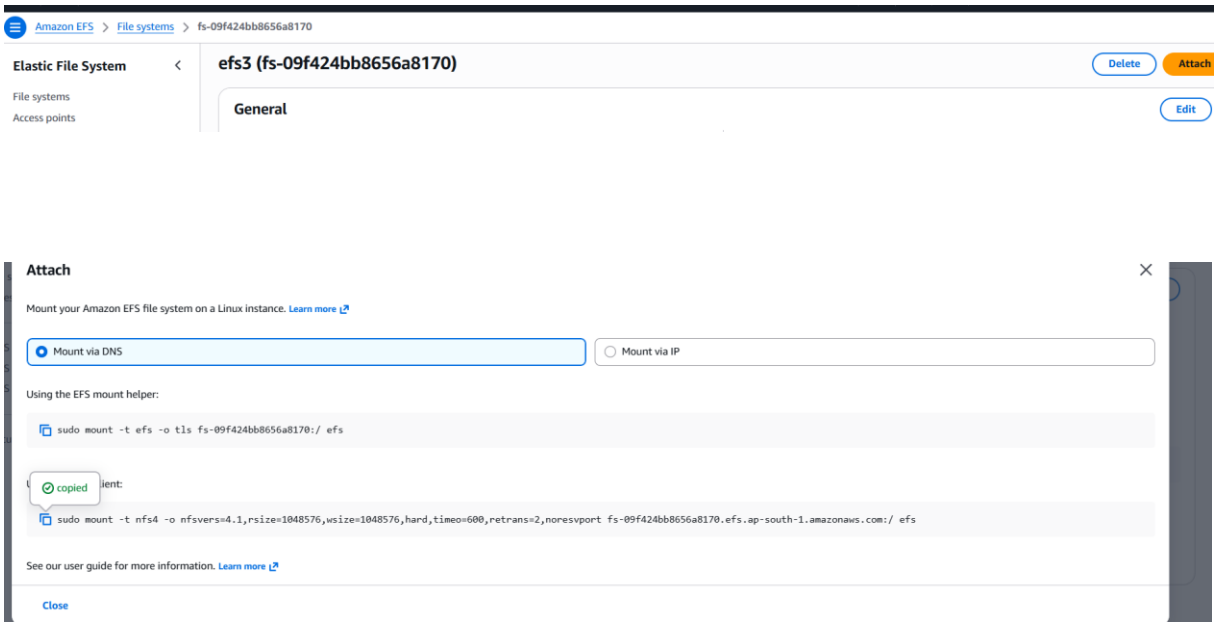
Screenshot: Connect to All EC2 Instances

1. Go to Amazon EC2 in the AWS Console.
2. Select the first instance.
3. Click Connect.
4. Repeat the same steps to connect all three instances.



Screenshot: Copy NFS Mount Link

1. Go to Amazon Elastic File System in the AWS Console.
2. Select the created EFS File System.
3. Click Attach (or Connect).
4. In the Mount via NFS section, copy the NFS mount link.



Screenshot: Mount EFS on EC2 Instances

1. Connect to Instance 1 in Amazon EC2.
2. Switch to root user:

   sudo su
3. Create a directory:

   mkdir jai
4. Paste and execute the NFS mount command copied from Amazon Elastic File System to mount the file system.
5. Repeat the same steps for Instance 2 and Instance 3

Screenshot: Create a File in Instance 1

1. Connect to Instance 1 in Amazon EC2.
2. Go to the mounted directory:

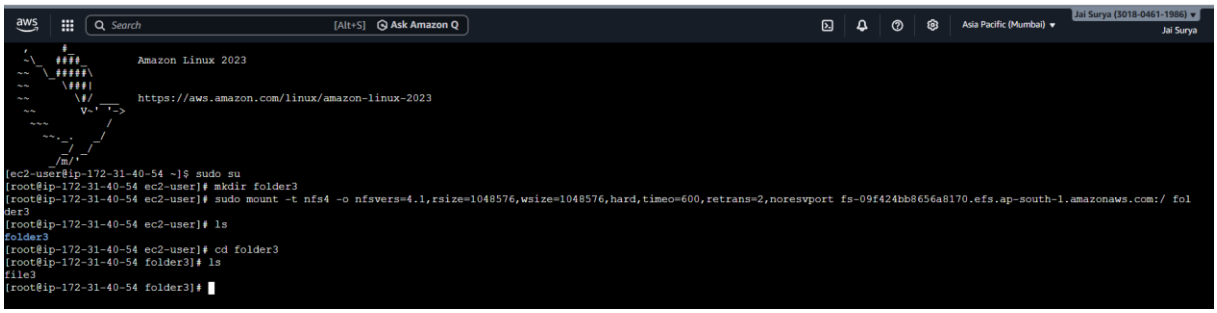
   cd jai
3. Create a file:

   touch file1.txt
4. Check the file:

   ls

Screenshot: File Visible in Instance 2 and 3

1. Connect to Instance 2 and 3 in EC2.
2. Navigate to the mounted directory:

   cd jai
3. List the files:

   ls
4. The file created in Instance 1 will be visible here because all instances are connected to the same Amazon Elastic File System.